

1 Three Dimensional Coordinate Systems and Vectors

1.7 Cylinders and Quadric Surfaces

Definition 1.7.1 A cylinder is a surface that consists of all lines that are parallel to a given line and pass through a given plane curve.

Definition 1.7.2 A quadric surface is the graph of a second-degree equation in three variables.

Example 1.7.3 Use traces to sketch the quadric surface with equation

$$x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1.$$

Example 1.7.4 Sketch the surface

$$z = 4x^2 + y^2.$$

Example 1.7.5 Sketch the surface

$$\frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1.$$

Example 1.7.6 Sketch the surface defined by the equation

$$4x^2 - y^2 + 2z^2 + 4 = 0.$$